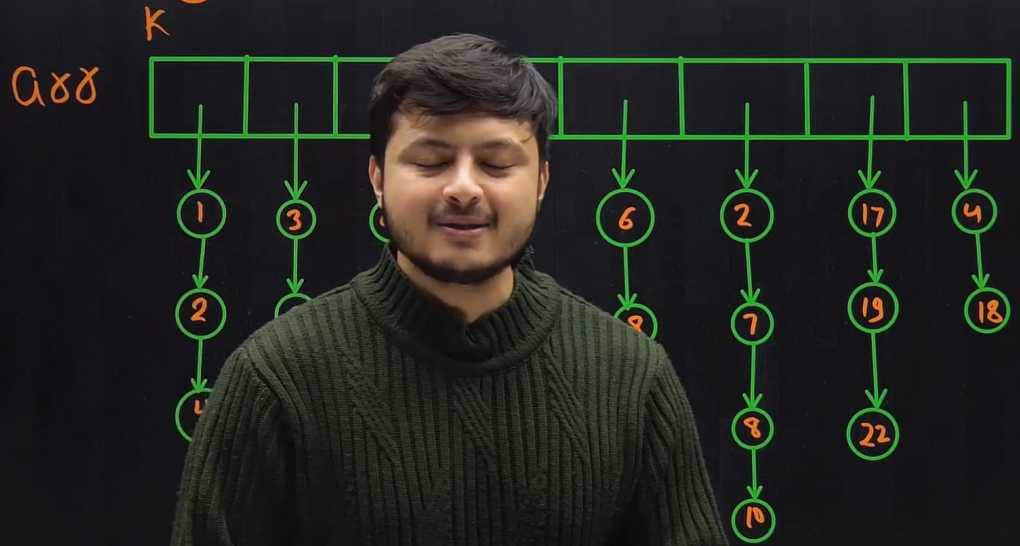
Merge K Sorted Linked List



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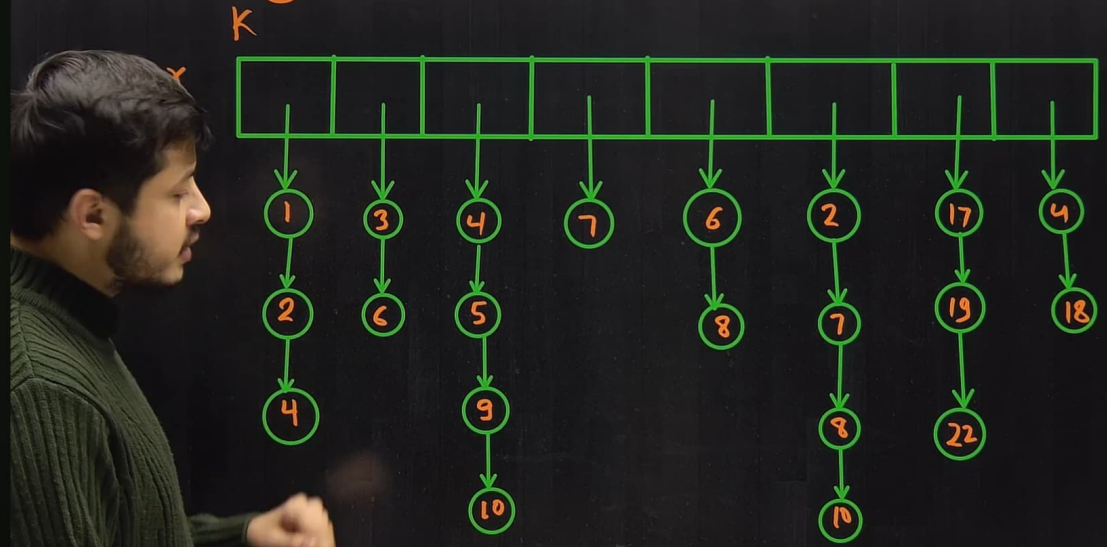

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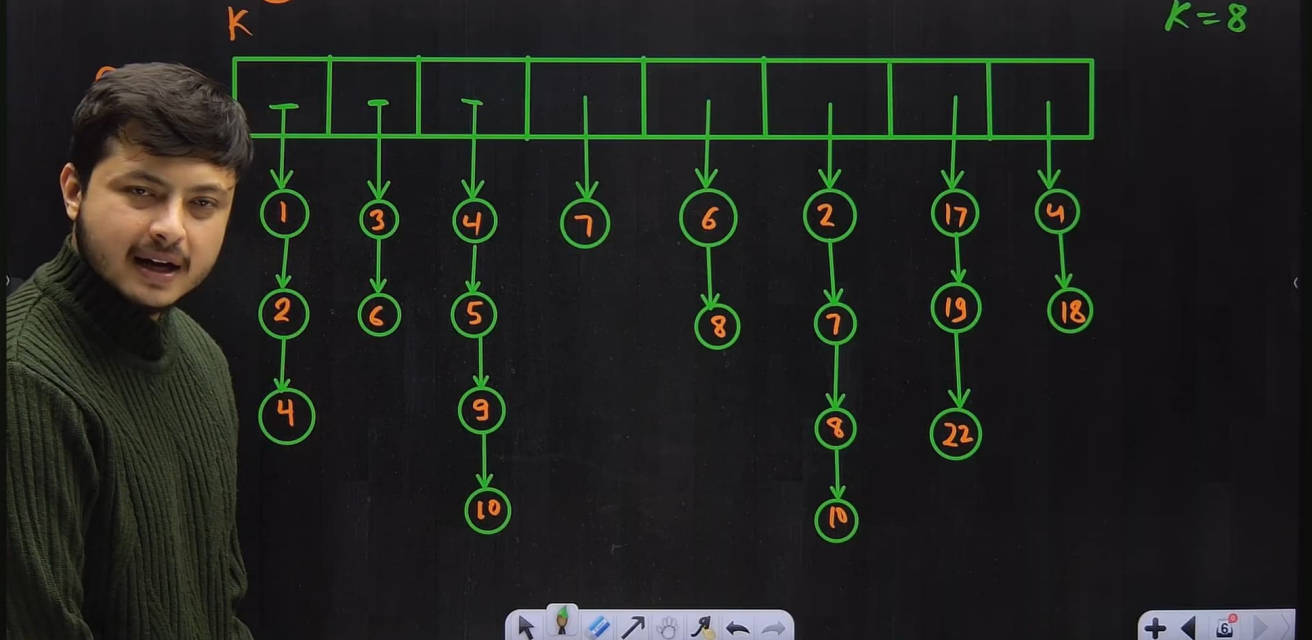

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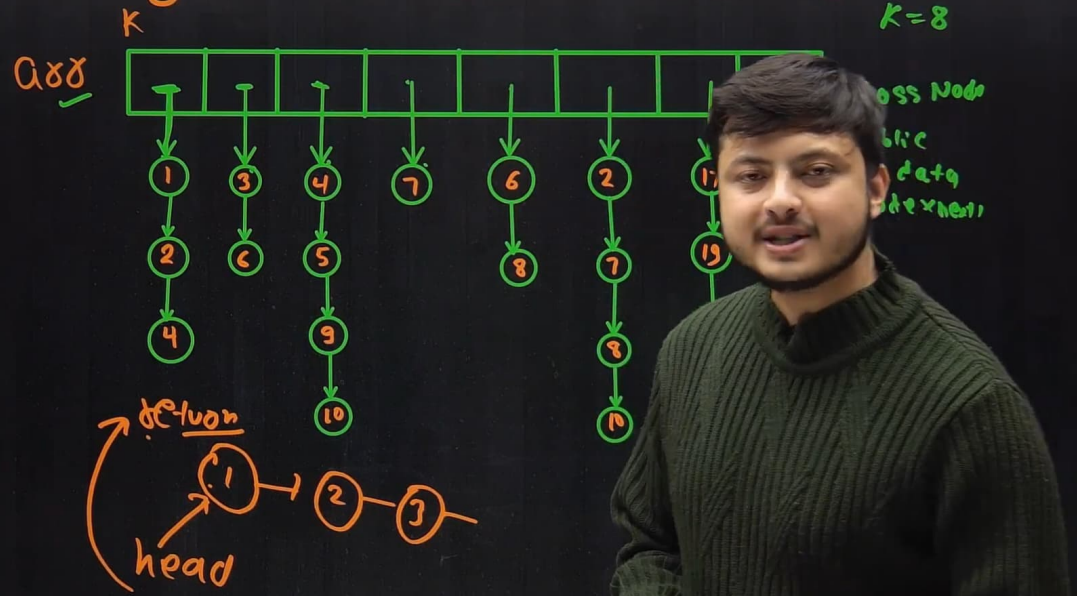

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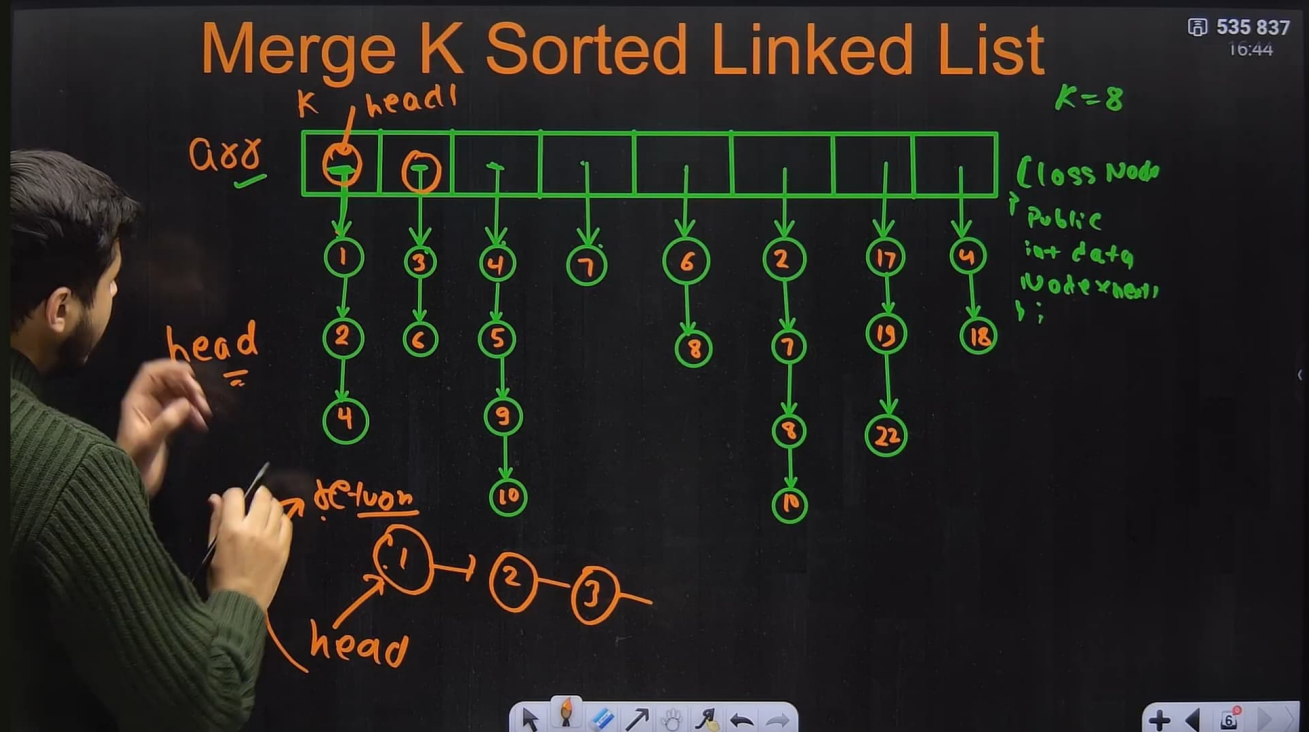
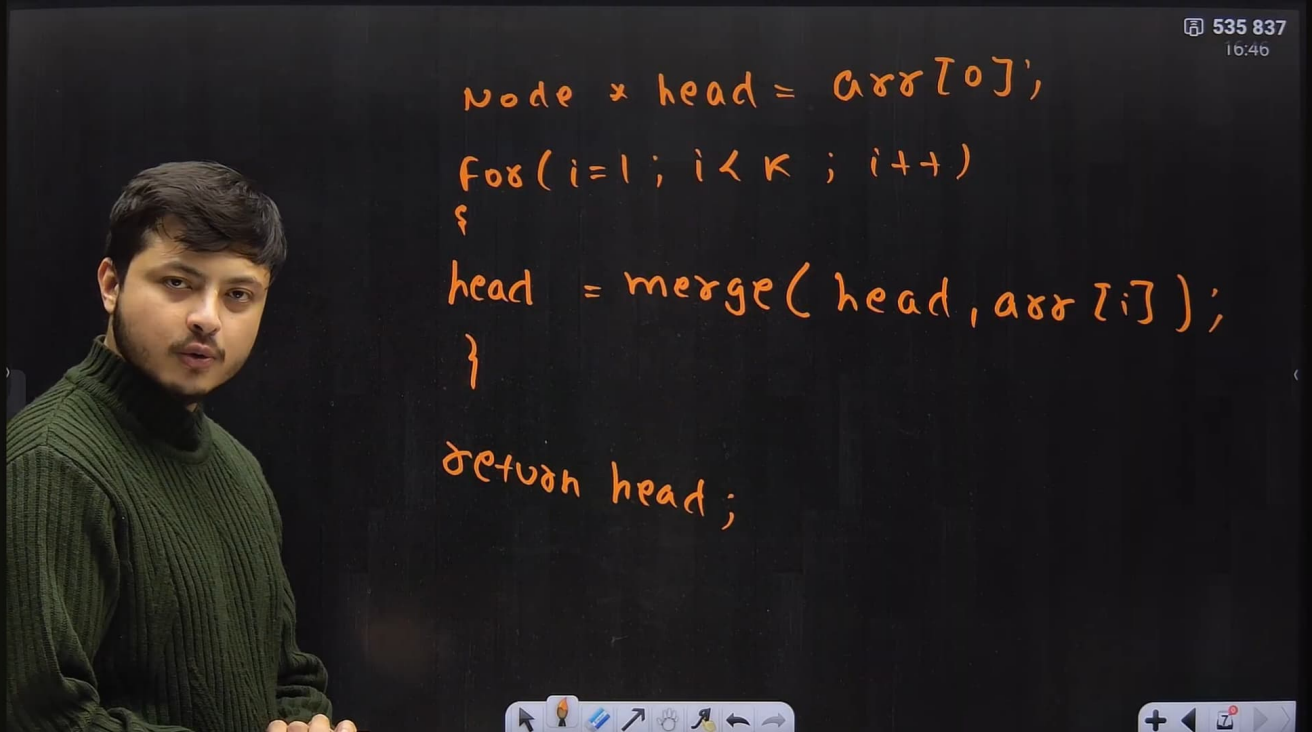
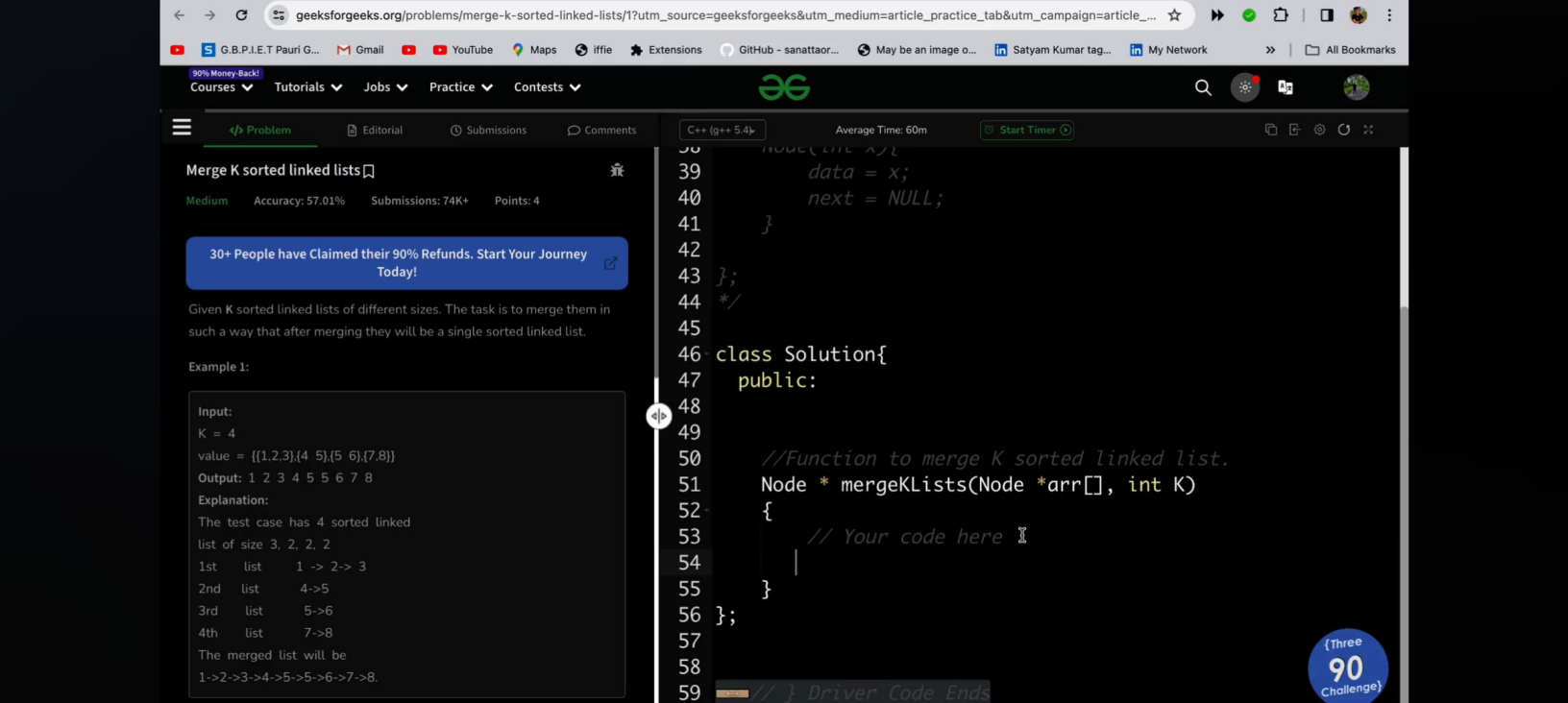



535 837
16:46
Node * head = arr[0];
For(i=1; i<K ; i++)
{
head = merge(head, arr[i]);
}
return head;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Average Time: 60m
Start Timer
Merge K sorted linked lists
Medium
Accuracy: 57.01%
Submissions: 74K+
Points: 4
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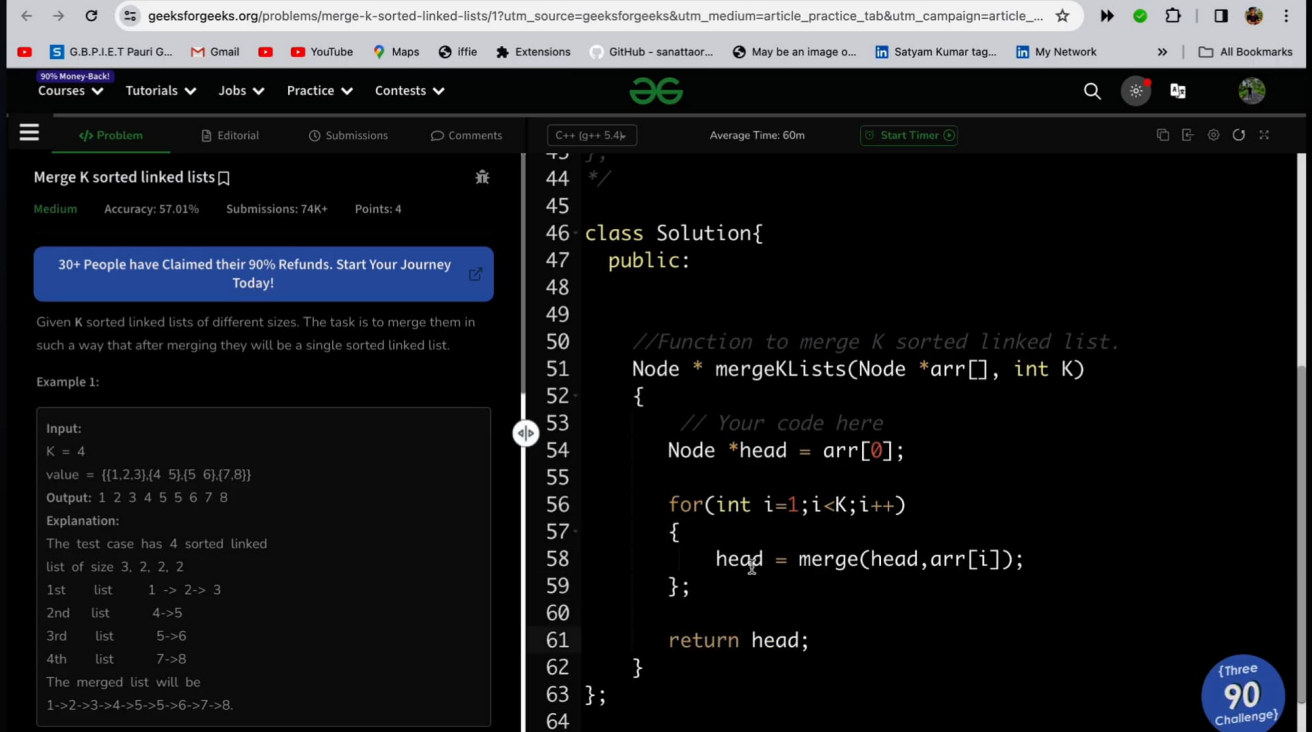
Given K sorted linked lists of different sizes. The task is to merge them in such a way that after merging they will be a single sorted linked list.
Example 1:
Input:
K = 4
value = {{1,2,3},{4 5},{5 6},{7,8}}
Output: 1 2 3 4 5 5 6 7 8
Explanation:
The test case has 4 sorted linked list of size 3, 2, 2, 2
1st list 1 -> 2-> 3
2nd list 4->5
3rd list 5->6
4th list 7->8
The merged list will be 1->2->3->4->5->5->6->7->8.
data = x;
next = NULL;
};
*/
class Solution{
public:
//Function to merge K sorted linked list.
Node * mergeKLists(Node *arr[], int K)
{
// Your code here
}
};
// } Driver Code Ends
{Three 90 Challenge}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*/
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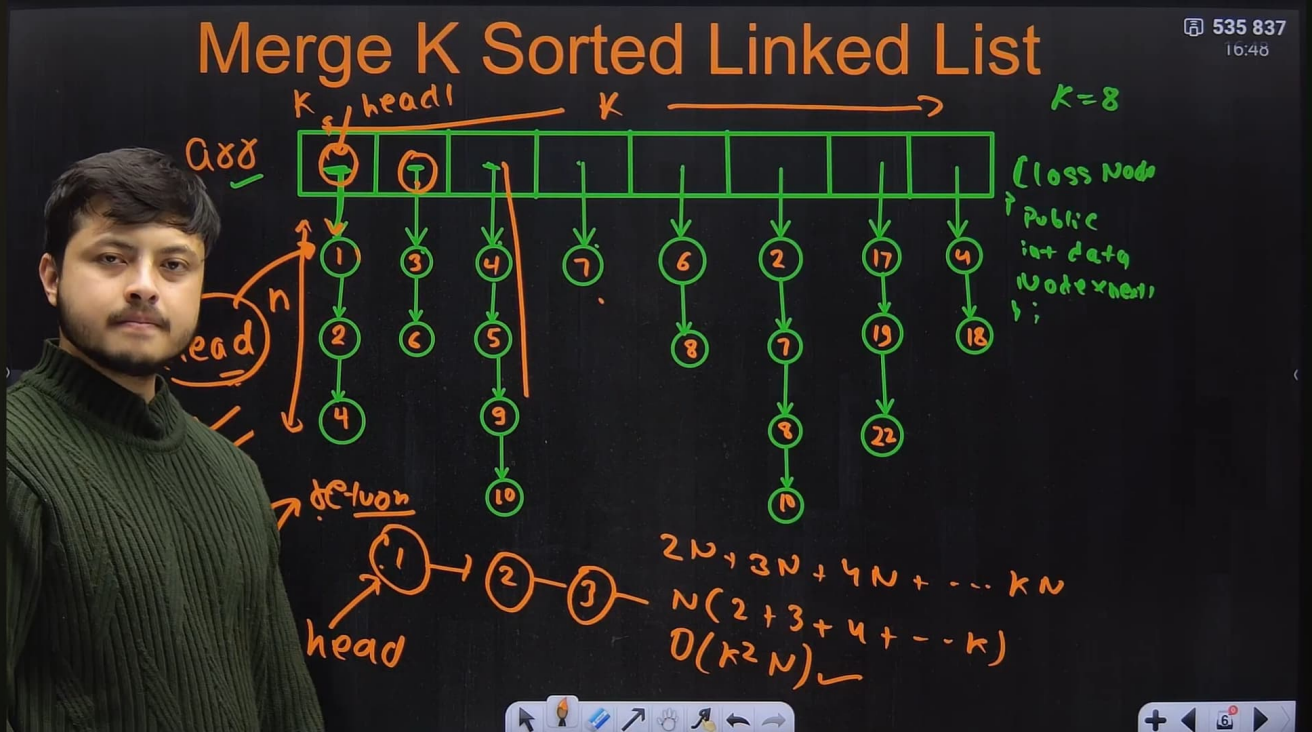
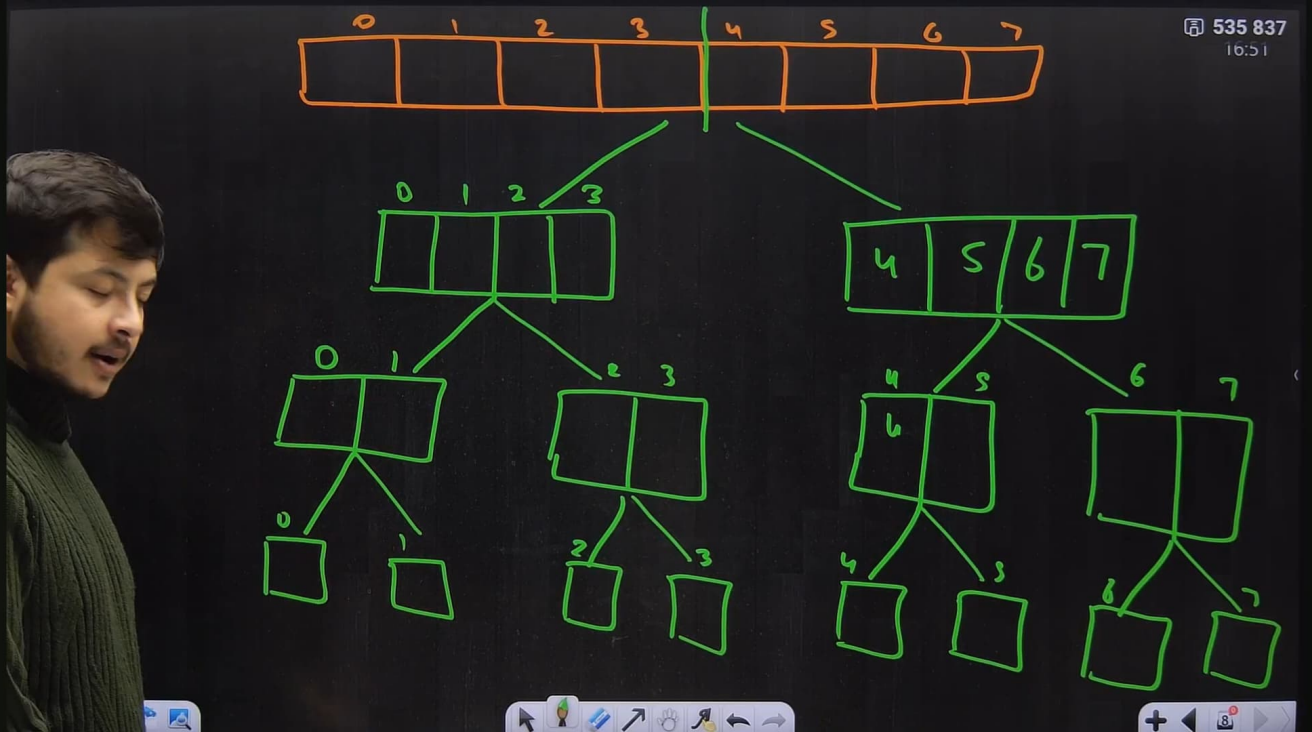
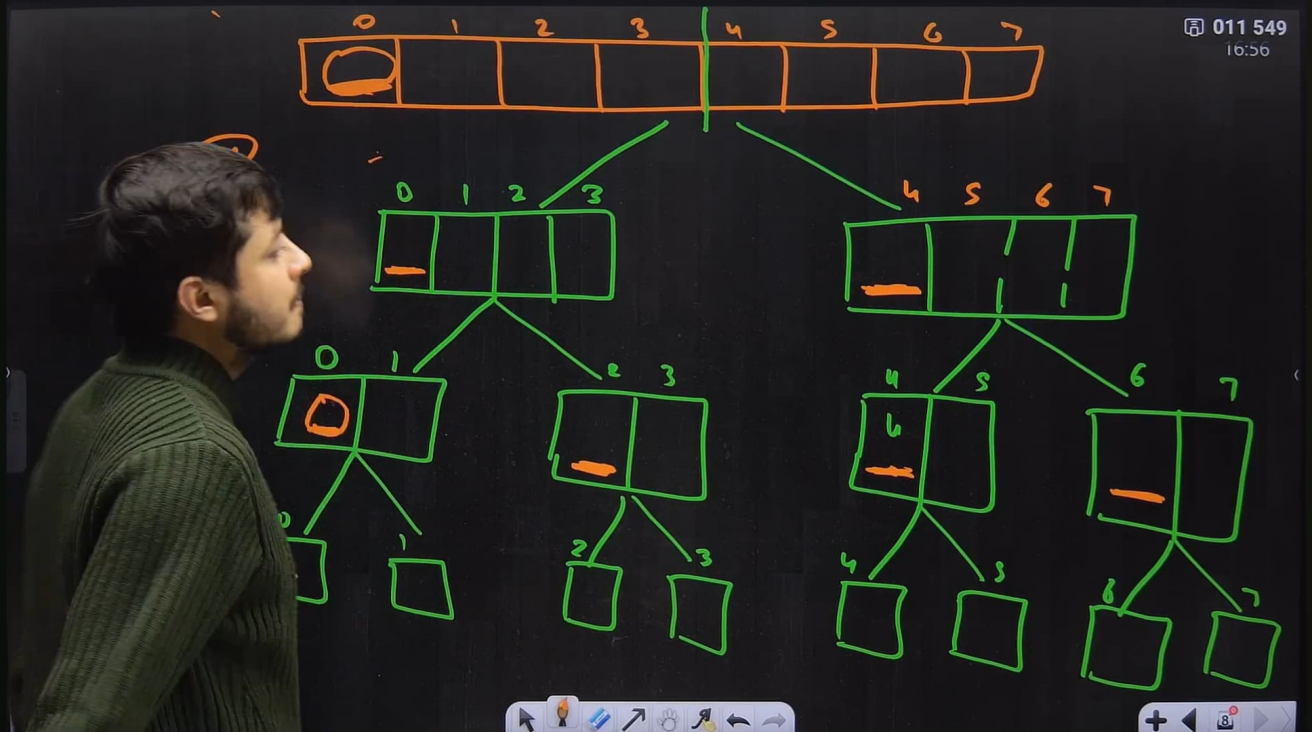
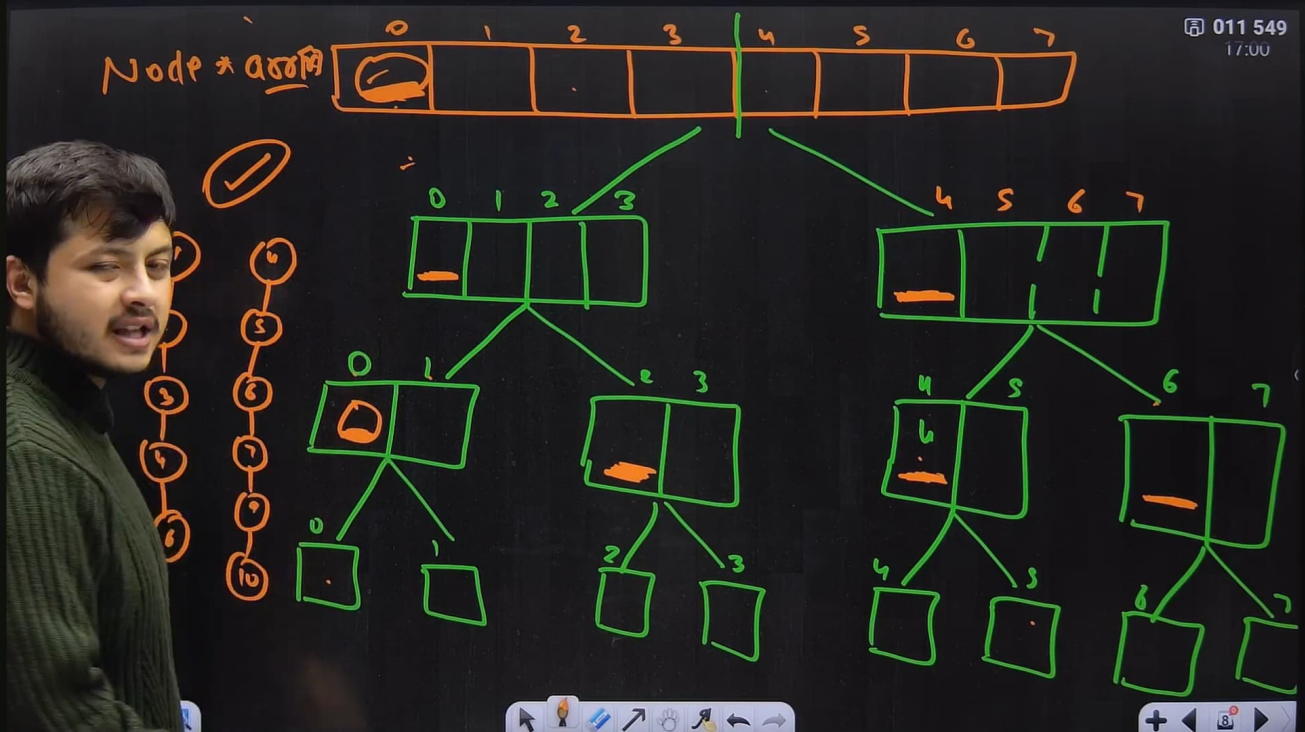
public:
//Function to merge K sorted linked list.
Node * mergeKLists(Node *arr[], int K)
{
// Your code here
Node *head = arr[0];
for(int i=1;i<K;i++)
{
head = merge(head,arr[i]);
};
return head;
}
};
{Three 90 Challenge}

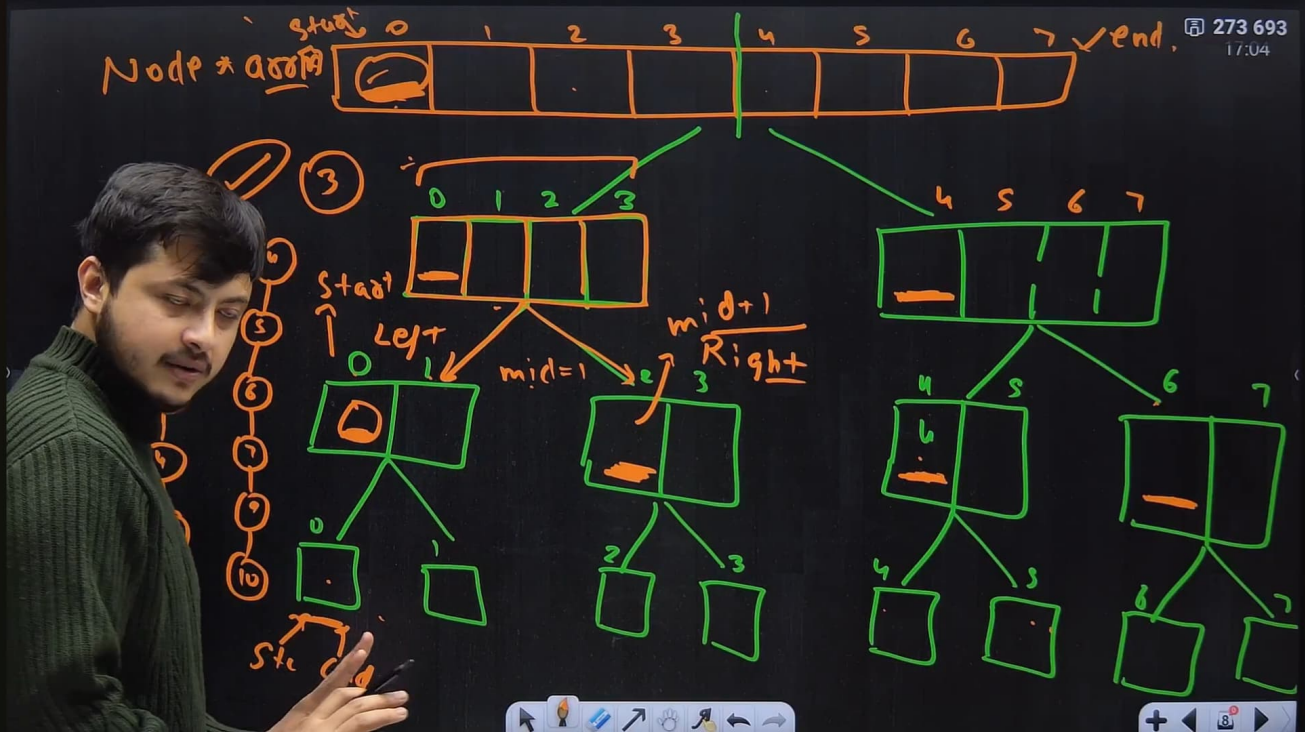
Merge K Sorted Linked List

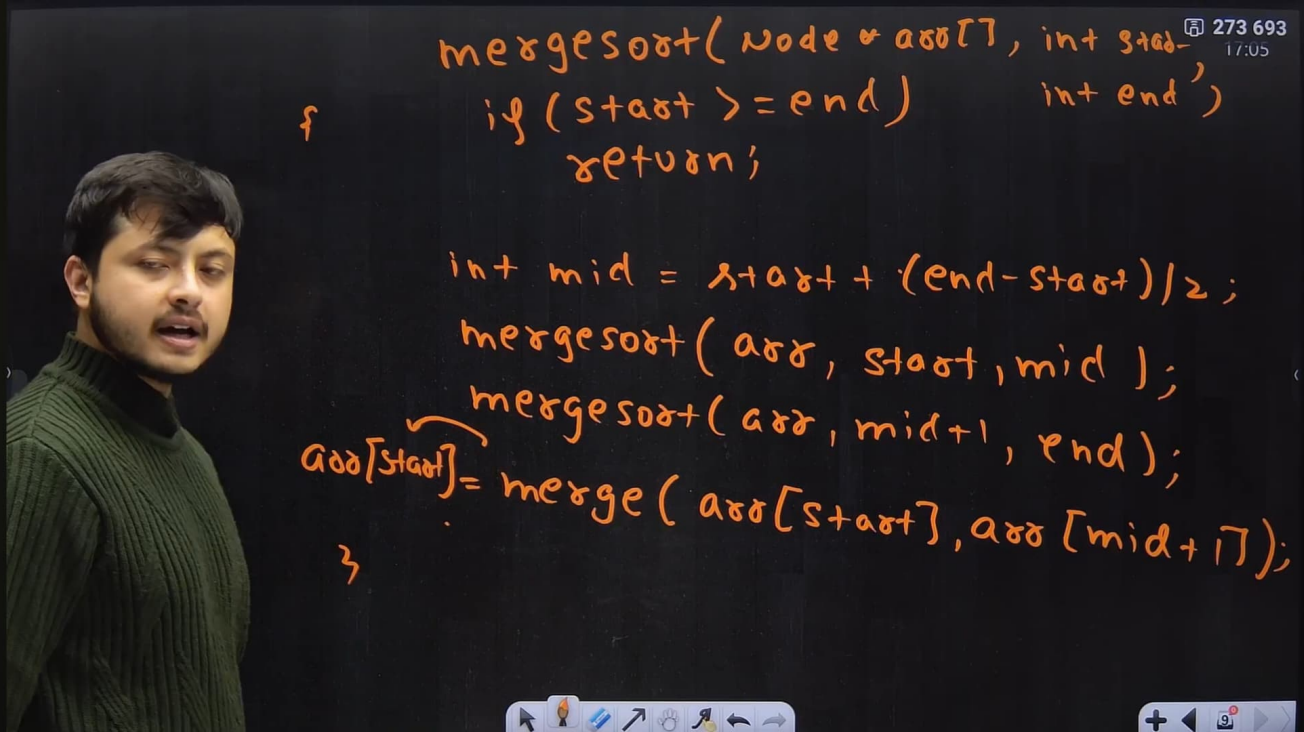


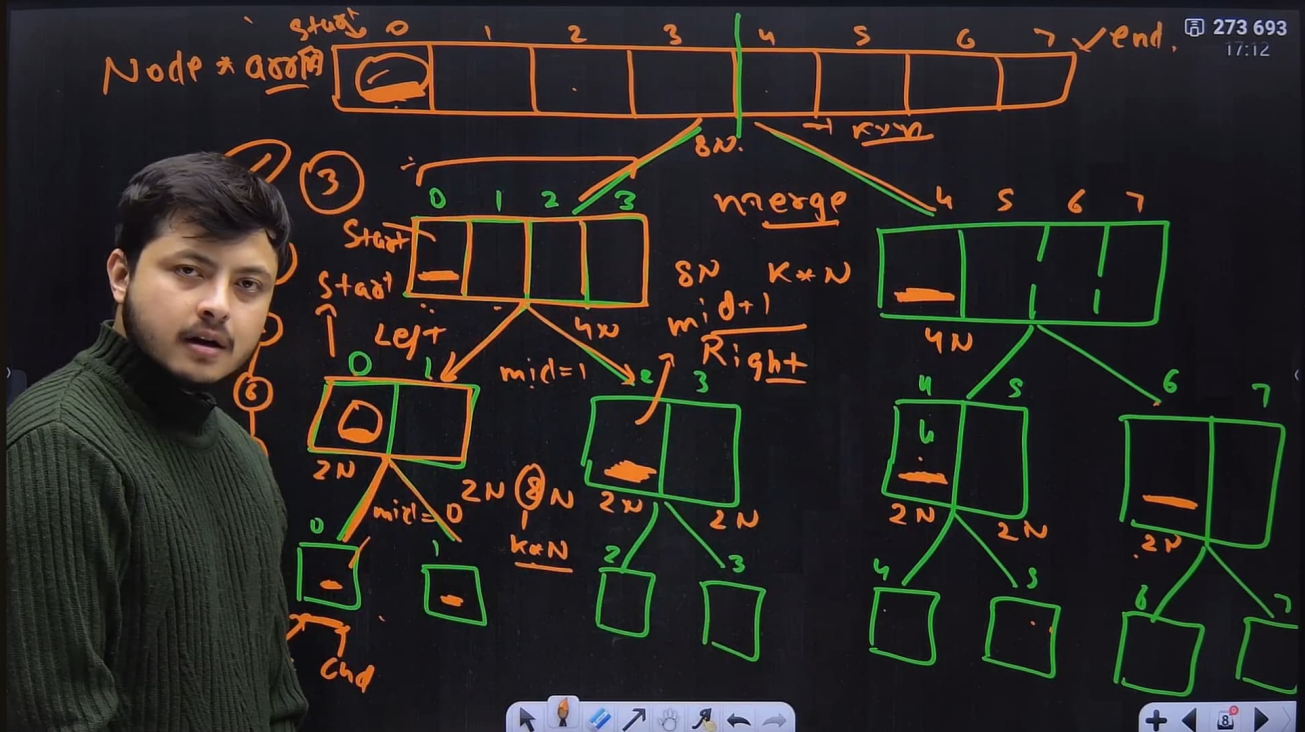

0 1 2 3 4 5 6 7
0 1 2 3
4 5 6 7
0 1
2 3
4 5
6 7
0
1
2
3
4
5
6
7
535 837
16:51

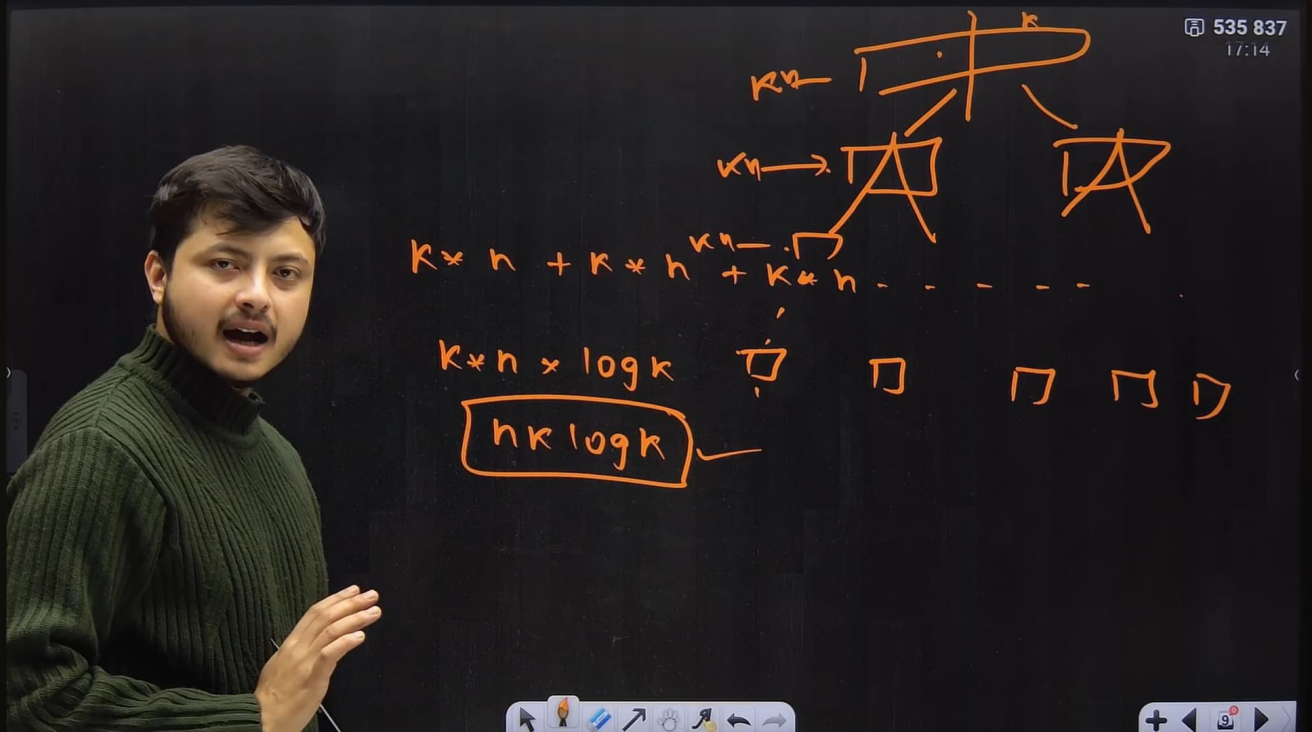

0 1 2 3 4 5 6 7
0 1 2 3
4 5 6 7
0 1
2 3
4 5
6 7


Node * arr[8]
0 1 2 3 4 5 6 7
011 549
17:00


Node * arr
start
end
0 1 2 3 4 5 6 7
mid=1
mid+1
Left
Right
start


mergesort(Node * arr[], int start, int end)
{
if (start >= end)
return;
int mid = start + (end-start)/2;
mergesort(arr, start, mid);
mergesort(arr, mid+1, end);
arr[start] = merge(arr[start], arr [mid+1]);
}
273 693
17:05


Node * arr
start
0 1 2 3 4 5 6 7
end
8N
K*N
merge
Start
Left
Right
mid=1
mid+1
4N
2N
mid=0
K*N
End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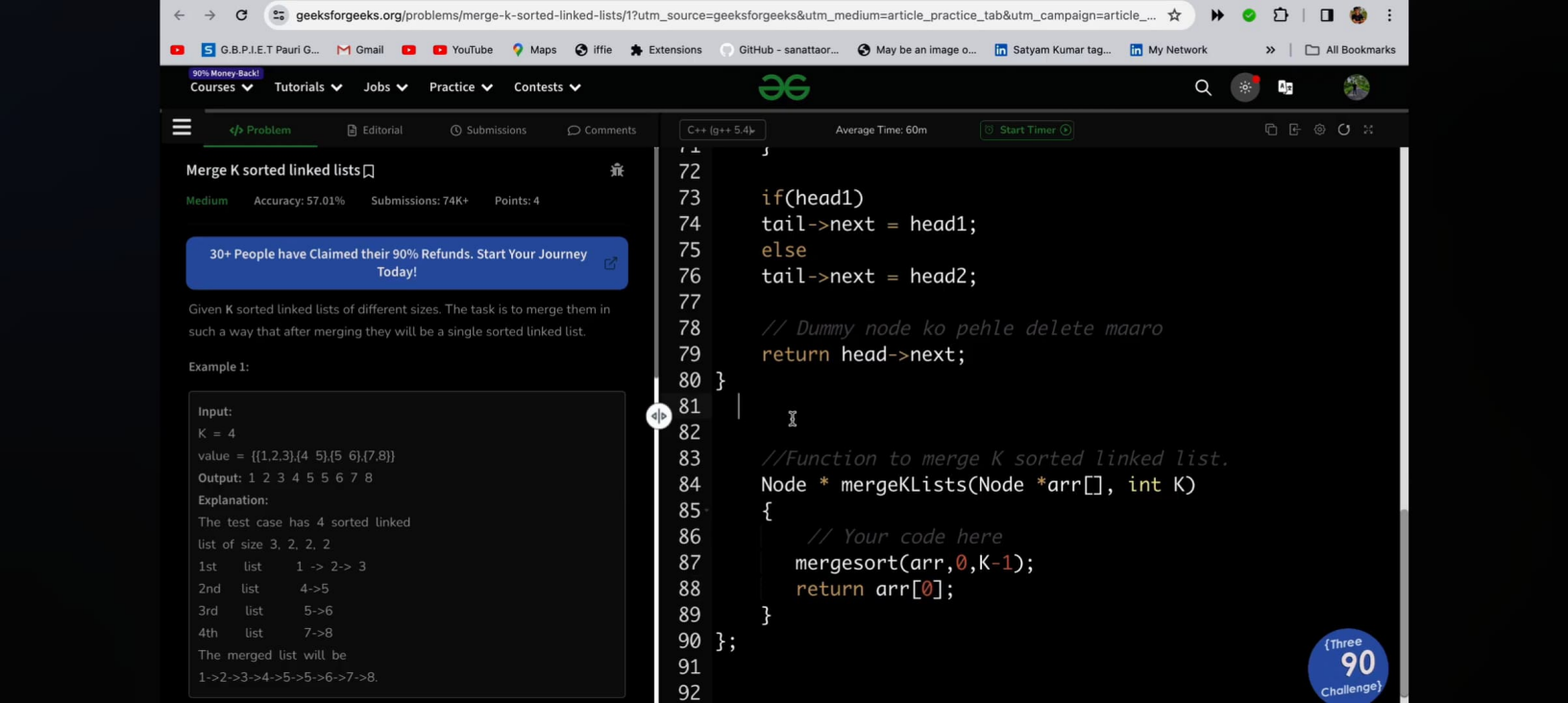
k
k*n
k*n
k*n
k*n + k*n + k*n - - - - - -
k*n * logk
hklogk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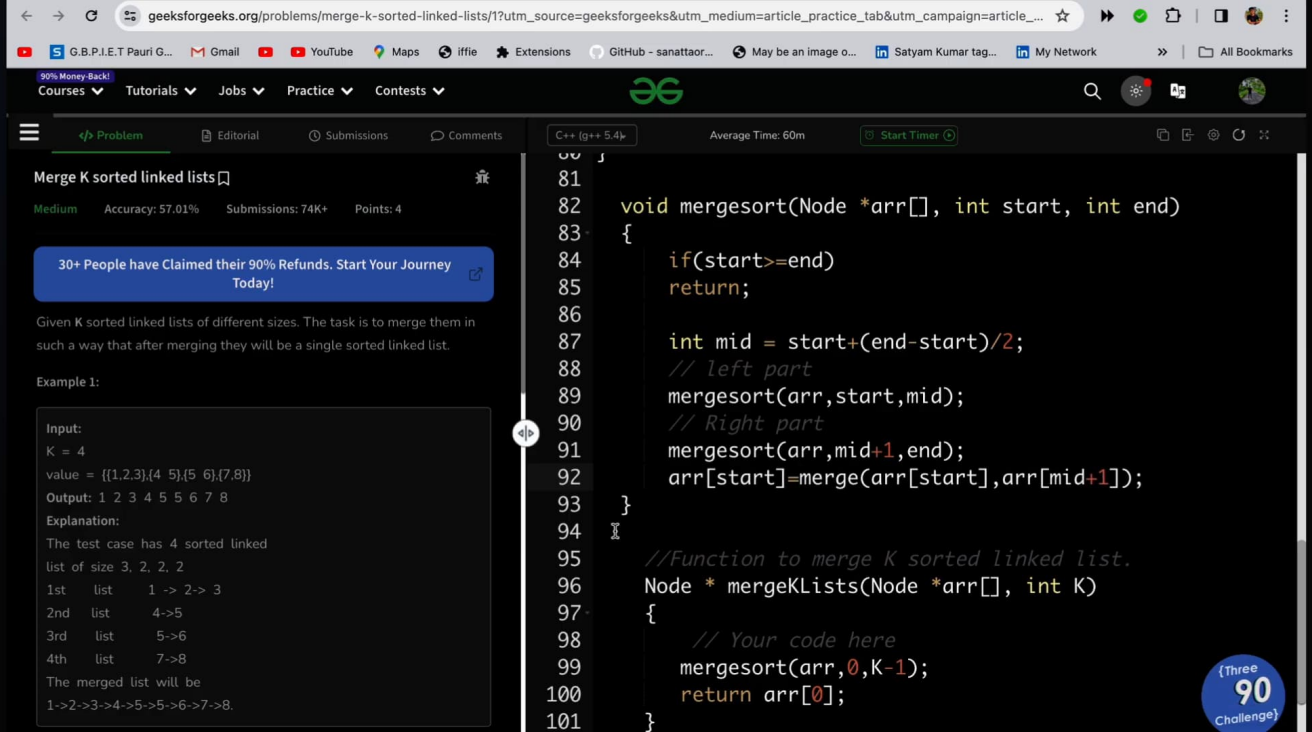
}
if(head1)
tail->next = head1;
else
tail->next = head2;
// Dummy node ko pehle delete maaro
return head->next;
}
//Function to merge K sorted linked list.
Node * mergeKLists(Node *arr[], int K)
{
// Your code here
mergesort(arr,0,K-1);
return arr[0];
}
};
{Three 90 Challenge}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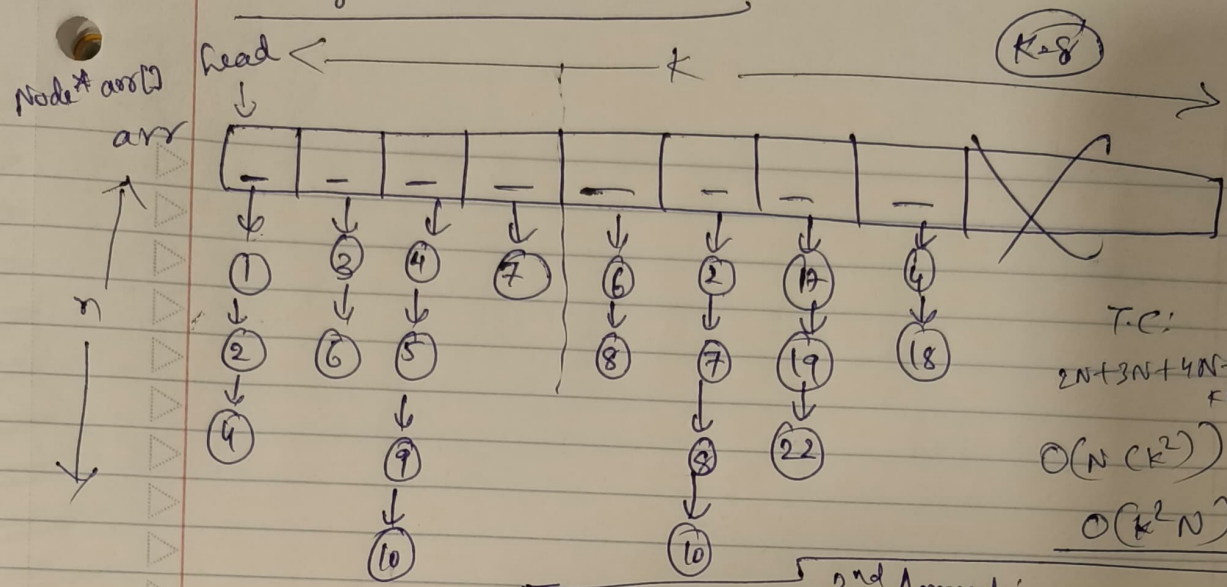
The merged list will be 1->2->3->4->5->5->6->7->8.
}
void mergesort(Node *arr[], int start, int end)
{
if(start>=end)
return;
int mid = start+(end-start)/2;
// left part
mergesort(arr,start,mid);
// Right part
mergesort(arr,mid+1,end);
arr[start]=merge(arr[start],arr[mid+1]);
}
//Function to merge K sorted linked list.
Node * mergeKLists(Node *arr[], int K)
{
// Your code here
mergesort(arr,0,K-1);
return arr[0];
}
{Three 90 Challenge}

Merge K Sorted LL



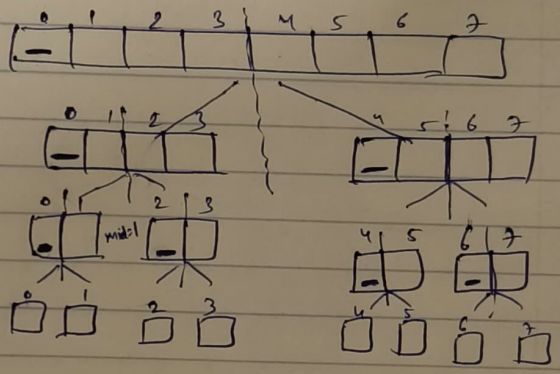
T.C:
$2N+3N+4N+\cdots+KN$
$O(N(K^2))$
$O(K^2N)$

Using similar/same Approach we followed to solve previous problem:

```
Node* head = arr[0]
for(int i=1; i<k; i++){
    head = merge(head, arr[i]);
}
return head;
```

Same as previous only instead of bottom we use *next

2nd Approach:




```
mergeSort(Node* arr[], int start, int end){
    if(start >= end)
        return
    int mid = start + (end-start)/2;
    mergeSort(arr, start, mid);
    mergeSort(arr, mid+1, end);
    arr[start] = merge(arr[start], arr[mid+1]);
}
```